

Anirudh Tyagi

✉ anirudhtyagi188@gmail.com

☎ +91 73032 89966

in [linkedin.com/in/itsanirudhtyagi](https://www.linkedin.com/in/itsanirudhtyagi)

🌐 [anirudh-tyagi](https://anirudh-tyagi.github.io)

Education

Sai University

Sep 2022 – May 2026

B.Tech in Computing and Data Science (CGPA - 7.7)

- **Relevant Coursework:** Machine Learning, Deep Learning, Reinforcement Learning, Artificial Intelligence, Data Structures & Algorithms, Speech and Image Analysis, Bioinformatics, Software Engineering, DBMS, Data Analysis and Visualization

Experience

Software Engineer Intern

Noida, India

Cadence Design Systems

Jan 2026 – Jun 2026

- Expanded into C++/Qt 5 product development for Allegro System Capture, an EDA application in the Allegro R&D team, using modern C++, STL, and OOP principles, implemented a real-time schematic world view widget with live viewport synchronization supported 9+ concurrent design workspaces, and an overhauled docking framework with deferred state restoration with 30 ms response time, orientation-aware resizing, and context-aware toolbar visibility improving multi-workspace behavior
- Designed a cryptographically secured expiry validation module for early-access builds using asymmetric digital signatures and authenticated encryption to enforce version-bound expiry controls, enforcing runtime validation of cryptographically signed license files; delivered <10ms end-to-end validation latency alongside a companion CLI for generating signed license artifacts

Software Engineer Trainee

Noida, India

Cadence Design Systems

Jun 2025 – Aug 2025

- Implemented Python/Flask-based ML data pipelines for Allegro R&D workflows, automating extraction and processing of 10k multi-level CCR archives, cleaning 40TB of schematic data, and managing structured datasets for model training and analytics
- Engineered connectivity-aware graph representations using BFS/DFS traversal for ML-based component placement, achieving 91% model accuracy, and integrated inference workflows with Allegro applications through socket-based communication for predictions in 100ms

Research Intern

Chennai, India

National Institute of Ocean Technology

Jul 2024 – Sep 2024

- Processed and analyzed 18 TB of deep-sea metagenomic data through bioinformatics pipeline of de novo assembly, BLAST homology search, and multi-sequence alignment, identifying candidate microbial genes with potential industrial biotechnology applications

Skills

Programming Languages: C++, C, Python, Julia, Tcl

Libraries & Frameworks: STL, Flask, Django, Node.js, Qt 5.0/6, PyQt, PyTorch, TensorFlow, Scikit-learn, OpenCV

Tools & Platforms: Git, Perforce, AWS, Docker, CMake, Jupyter Notebook, Jira, RHEL

AI/ML: Machine Learning, Deep Learning, Reinforcement Learning, Feature Engineering, Model Evaluation, Data Processing

Projects

Smart Home Energy AI

[smart-home-ml-rl](#) 

Python, PyTorch, Scikit-learn, Reinforcement Learning, Flask, Gymnasium, SQLAlchemy

- Developed a reinforcement learning-based energy optimization system integrating 3 supervised ML models within a Gymnasium environment, achieving 26.4% simulated energy reduction, 98% comfort rating, and 97.8% occupancy prediction accuracy
- Built a Flask/SocketIO dashboard synchronizing 50+ IoT streams with the optimization engine, achieving sub-15 ms inference latency and 200 ms end-to-end response time

Skyward

[/anirudh-tyagi/Game-cpp-qt](#) 

C++, Qt 6/QML, CMake

- Designed the collision pipeline of a 60 Hz fixed-timestep game loop around a uniform-grid spatial hash, benchmarked against an exhaustive $O(n \times m)$ sweep: $72\times$ fewer per-frame AABB pair tests (250,613 to 3,497) and $8.7\times$ lower collision time (4.0 ms to 0.46 ms) at 1,000-bullet $\times$ 2,000-enemy load; verified pair-for-pair across 2,000 randomized trials and profiled the crossover at 100×250 entities

Research & Publications

- Behl D., Tyagi A., Kandhoul N., Dhurandher S. “Post Quantum Federated learning and Blockchain with Resilient IPFS for Cyber logs (FABRIC)”, under review at *Journal of Network and Systems Management*, Springer.
- Tyagi A., Behl D. “Optimized Routing via Blockchain and Intelligent Edge-Cloud Architecture (ORBIT)”, under review at *Transactions on Network and Service Management*, IEEE.

Leadership & Activities

Cadence India Hackathon

2025

- Presented BLINC, a decentralized log analysis system integrating federated learning, blockchain (PoS), and IPFS

Sports Society President, Sai University

2022-2023

- President, Sports Society. Managed student activities and coordinated internal events